Mohammad Ayaz Alam

(+49) 017684051947 | alam.ayaz47@gmail.com | github.com/rebel47 | linkedin.com/in/ayaz-alam

SKILLS

Generative AI Systems:	Agentic AI, LangGraph, LangChain, LangSmith (Tracing & Evaluation), MCP, RAG
AI Development Tools:	Cursor, Claude Code, GitHub Copilot, OpenAI Codex
Natural Language Processing:	Fine-tuning, Prompt Engineering, Hugging Face, Transformers
Machine Learning:	Supervised & Unsupervised Learning, Feature Engineering, Model Evaluation, PyTorch
Programming:	Python, SQL, JavaScript
Backend & Web Development:	FastAPI, Django, Flask, Next.js, REST APIs
Cloud & DevOps:	AWS (EC2, S3, Lightsail), Docker, Git/GitHub, CI/CD, Linux
Databases:	PostgreSQL, Firebase, Query Optimization
Data & Visualization:	Pandas, NumPy, Scikit-learn, Power BI, Plotly
Other:	Flutter, Firebase, HTML, CSS, LaTeX

EXPERIENCE

Zalion GmbH, *Junior AI Engineer* | Munich, Germany — Mar 2026 – Jul 2026

- Developed multi-agent procurement workflows using LangGraph and specialized AI agents.
- Integrated agent endpoints into chat workflows for task-specific AI interactions.
- Built backend APIs using FastAPI and frontend components using Next.js.
- Optimized agent orchestration and execution latency to improve response times.
- Contributed to AWS EC2 deployment, Docker infrastructure, and CI/CD pipelines.

Siemens Energy, *Master's Thesis – Junior Data Scientist (NLP)* | Erlangen, Germany — Jul 2025 – Feb 2026

- Designed a modular multi-agent framework using LangGraph and LangChain for industrial NLP applications.
- Implemented agent orchestration, tool execution, and persistent conversation state.
- Built FastAPI endpoints with streaming responses and OpenAI-compatible interfaces.
- Integrated PostgreSQL checkpointing and execution audit trails for reproducible, traceable workflows.
- Applied the framework to two application Factory Assistance workflow and Terminal Management System.
- Implemented human-in-the-loop approvals for external actions such as email sending.

Ilham Labs.ai, *Founder & Product Engineer* | Independent Product Initiative, Germany — 2025 – Present

- Owned architecture, full-stack development, deployment, and security across independent products.
- Built Lattice Lab, a visual multi-model AI workspace with parallel inference, branching workflows, and context merging.
- Implemented graph-based conversations, authenticated workspaces, and PostgreSQL persistence for Lattice Lab.
- Developed Air Lune, a multiplayer focus platform using FastAPI, Next.js, and LiveKit for real-time collaboration.
- Built WhenYouOpen.me, a scheduled message platform using FastAPI, AWS S3, Docker, and Nginx.

Institute for Employment Research (IAB), *Research Assistant* | Nuremberg, Germany — Nov 2024 – Feb 2026

- Applied XGBoost, Random Forest, SVM, and positive-unlabeled learning to labor-market analysis and job-offer ranking.
- Built scalable preprocessing and feature-engineering pipelines for datasets containing 90M+ records.
- Developed predictive models to rank job offers and support caseworker decision-making.
- Addressed class imbalance using weight estimation and alternative positive-unlabeled learning strategies.
- Conducted exploratory data analysis, model evaluation, visualization, and research reporting.

BASF, *Data Consultant Intern* | Ludwigshafen am Rhein, Germany — Apr 2025 – Jul 2025

- Developed Generative AI tools to support internal business workflows.
- Built a data-validation AI agent for structured-data checks, configurable tests, and quality scoring.
- Implemented automated validation reports to communicate data-quality issues and results.
- Developed an internal event application covering submissions, voting, analytics, and access controls.
- Contributed to internal digital tools for project management and operational workflows.

PROJECTS

Ilmora | Offline Android LLM Application

- Built an offline LLM application in Flutter for on-device inference with GGUF models.
- Integrated llama.cpp with Android NDK and CMake for local model execution.

Fine-Tuned GPT-2 for Poetry Generation with Reinforcement Learning

- Fine-tuned GPT-2 (124M) on a poetry corpus using Hugging Face Transformers.
- Applied custom reward functions for rhyme, coherence, and creativity to guide generation.
- Experimented with policy-gradient optimization to improve poetry generation quality.

CERTIFICATES

Huggingface - AI Agents Fundamentals - Agents fundamentals, Tools and Action, Agents Workflow
iNeuron.ai - Full Stack Data Analytics - Python, SQL, Snowflake, PowerBI, DAX, Excel
iNeuron.ai - Deep Learning Foundation - Neural Network, Backpropagation, CNN
Kaggle - Intro to Machine Learning - Model Validation, Underfitting-Overfitting, Random Forest
freeCodeCamp - Data Analysis with Python - Numpy, Pandas, Data Cleaning

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg, *MSc. Data Science* | Erlangen, Germany — 2022 - 2026
- Courses: Business Analytics, ML Time Series, ML in Finance, Mathematics of Learning, Linked Data

Guru Gobind Singh Indraprastha University, *Bachelors of Computer Applications* | Delhi, India — 2018 - 2021
- Courses: OOPs, Network Security, DBMS, Web Technologies, Data Warehousing and Data Mining

Languages

English	Professional proficiency
German	A2 proficiency
Hindi & Urdu	Native proficiency